

DEFENCE RESEARCH & ELECTRONICS TESTING LABORATORY

NOTICE INVITING TENDER: DETL/PROC/LAB-PS/2026/15

TENDER FOR: Supply of High-Precision Programmable Laboratory DC Power Supplies (0-60V, 0-30A, 900W)

SECTION 3: TECHNICAL SPECIFICATIONS FOR LABORATORY DC POWER SUPPLY

1. OUTPUT RATINGS & PERFORMANCE:

- 1.1 Output Voltage: Continuously programmable from 0 to 60 Volts DC.
- 1.2 Output Current: Continuously programmable from 0 to 30 Amperes DC.
- 1.3 Output Power: Total continuous output power shall be minimum 900 Watts with autoranging V/I envelope.
- 1.4 Line Regulation:  $\leq 0.01\% + 2 \text{ mV}$  for voltage;  $\leq 0.01\% + 250 \text{ uA}$  for current.
- 1.5 Load Regulation:  $\leq 0.01\% + 2 \text{ mV}$  for voltage;  $\leq 0.01\% + 250 \text{ uA}$  for current.
- 1.6 Voltage Ripple and Noise (20 Hz to 20 MHz): Not exceeding 2 mVrms / 20 mV peak-to-peak.
- 1.7 Current Ripple: Less than 5 mArms.

2. ACCURACY & PROGRAMMING RESOLUTION:

- 2.1 Programming Resolution: Voltage resolution  $\leq 1 \text{ mV}$ ; Current resolution  $\leq 1 \text{ mA}$ .
- 2.2 Readback Accuracy: Voltage  $\pm (0.05\% + 10 \text{ mV})$ ; Current  $\pm (0.1\% + 10 \text{ mA})$ .
- 2.3 Transient Response Time: Less than 50 microseconds for output recovery to within 15 mV following a 50% to 100% load step.

3. OPERATING MODES & PROTECTIONS:

- 3.1 Seamless automatic crossover between Constant Voltage (CV) and Constant Current (CC) modes.
- 3.2 Comprehensive hardware and software protection: Over Voltage Protection (OVP), Over Current Protection (OCP), Over Temperature Protection (OTP).
- 3.3 Output enable/disable electronic soft switch with isolated floating output terminals.

4. USER INTERFACE & REMOTE CONNECTIVITY:

- 4.1 Minimum 4.3-inch high-resolution color TFT display showing simultaneous real-time voltage, current, power, and operational mode.
- 4.2 Standard remote programming interfaces: USB (USBTMC), RS-232, and LAN (LXI Core 2011 compliant) with SCPI command set.
- 4.3 Analog programming and monitoring interface (0-5V / 0-10V control inputs).

5. ELECTRICAL & ENVIRONMENTAL SPECIFICATIONS:

- 5.1 Universal AC input: 100V to 240V AC ( $\pm 10\%$ ), single phase, 47-63 Hz with active Power Factor Correction (PFC  $\geq 0.98$ ).
- 5.2 Operating temperature:  $0^{\circ}\text{C}$  to  $40^{\circ}\text{C}$ , Relative humidity: up to 80% non-condensing.
- 5.3 Cooling: Forced air cooling with low-noise temperature-controlled smart fan.
- 5.4 Chassis earthing terminal complying with IS 3043 Code of Practice for Earthing.

6. SAFETY & QUALITY STANDARDS:

- 6.1 Safety conformity: Conformance to IS/IEC 61010-1 Safety Requirements for Electrical Equipment for Measurement, Control, and Laboratory Use.
- 6.2 Electromagnetic Compatibility (EMC): Conformance to IS/IEC 61326-1 for laboratory equipment and IS/IEC 61000-3-2 for AC harmonic emission limits.
- 6.3 Calibration: Valid calibration certificate from an NABL accredited laboratory (ISO/IEC 17025) with reported measurement uncertainties.
- 6.4 3 Years Comprehensive OEM warranty including parts and calibration support.